Strict log-concavity of the gaps between squared Bessel zeros

A solution to MathDB 326635

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Abstract

The ratios of consecutive gaps between the squared positive zeros of J_1, with the additional zero $j_{1,0} = 0$, are strictly decreasing. This proves Conjecture 3 of Brix, Canuto, and Dahmen. The proof uses an affine comparison for the equation $u'' + u/(4z) = 0$ and includes the singular endpoint at zero.

Task 1: solve the original problem

Let $j_{1,k}$ be the kth positive zero of J_1, set $j_{1,0} = 0$, and put

$$\lambda_k = j_{1,k}^2, \qquad d_k = \lambda_k - \lambda_{k-1}, \qquad q_k = \frac{d_{k+1}}{d_k} \quad (k \geq 1).$$

Brix–Canuto–Dahmen [1], Conjecture 3, ask whether q_k decreases. The same question is explicitly retained as Conjecture 3.36, equation (3.51), in Brix's later thesis [2], page 38.

Theorem 1. *For every $k \geq 1$,*

$$d_{k+1} > d_k > 0, \qquad d_{k+1}^2 > d_k d_{k+2}.$$

Equivalently,

$$q_1 > q_2 > q_3 > \cdots > 1.$$

Moreover, $q_k \to 1$, and the sharp upper bound on the sequence is $q_1 = j_{1,2}^2/j_{1,1}^2 - 1$.

The equation for the squared zeros

The entire function

$$u(z) = 2\sqrt{z}\, J_1(\sqrt{z}) = z \sum_{n=0}^{\infty} \frac{(-z/4)^n}{n!(n+1)!}$$

is real on the real axis and satisfies

$$u''(z) + \frac{u(z)}{4z} = 0 \qquad (z > 0). \tag{1}$$

This follows directly from its power series or the Bessel equation. Its nonnegative zeros are exactly $\lambda_0, \lambda_1, \ldots$, all simple; in particular

$$u(z) = z + O(z^2), \qquad u'(0) = 1.$$

Simplicity at a positive zero also follows from uniqueness for (1).

Comparing two successive intervals

Fix $k \geq 1$, and write

$$a = \lambda_{k-1}, \quad b = \lambda_k, \quad c = \lambda_{k+1}, \quad d = \lambda_{k+2}, \qquad r = \frac{c-b}{b-a}, \quad s = b - ra.$$

Thus $r > 0$, and the affine function $T(z) = rz + s$ sends a to b and b to c. Multiply u by a sign, if necessary, so that it is positive on (a, b). Since its signs alternate at consecutive zeros,

$$v(z) = -u(T(z))$$

is also positive on (a, b), and both u and v vanish at its endpoints. The function v solves

$$v''(z) + \frac{r^2}{4(rz+s)} v(z) = 0.$$

Set

$$D(z) = \frac{r^2}{4(rz+s)} - \frac{1}{4z} = \frac{r(r-1)z - s}{4z(rz+s)}. \tag{2}$$

For $z > a$ the denominator is positive. With $W(z) = u'(z)v(z) - u(z)v'(z)$, differentiation gives

$$W'(z) = D(z)u(z)v(z). \tag{3}$$

Both endpoint Wronskians vanish, hence

$$\int_a^b D(z)u(z)v(z)\, dz = 0. \tag{4}$$

When $a = 0$, this is an improper integral justified without any regular endpoint assumption: $u(z) = O(z)$, $v(z) = O(z)$, their derivatives are bounded, and $D(z) = O(1/z)$. Thus the integrand is $O(z)$ and $W(z) \to 0$ as $z \downarrow 0$.

If $r \leq 1$, then $s = b - ra \geq b - a > 0$, so the numerator in (2) is strictly negative throughout (a, b). This contradicts (4), because $uv > 0$ there. Therefore $r > 1$. The numerator of (2) is now a strictly increasing affine function. Its integral against the positive weight $uv/[4z(rz+s)]$ is zero, so it must change sign at a point strictly between a and b. In particular,

$$D(z) > 0 \qquad (z \geq b). \tag{5}$$

The next zero of v after b is $e = T^{-1}(d)$. We claim $e < c$. Otherwise u and v are both negative on (b, c). Equations (3) and (5) give $W(c) > W(b) = 0$. But $u(c) = 0$, $u'(c) > 0$, and $v(c) \leq 0$, so

$$W(c) = u'(c)v(c) \leq 0,$$

a contradiction. Hence $e < c$, or $d < T(c)$. Since $T(b) = c$, this is

$$d - c < r(c - b).$$

It follows that

$$1 < r = \frac{c-b}{b-a}, \qquad \frac{d-c}{c-b} < \frac{c-b}{b-a},$$

which proves both strict inequalities in Theorem 1, for every k.

The limit and the sharp bound

The decreasing sequence $q_k > 1$ has a limit $L \geq 1$. If $L > 1$, then $d_{k+1} \geq L d_k$, so the gaps, and consequently λ_k, grow at least geometrically. This contradicts the classical Bessel zero asymptotic $j_{1,k}/k \to \pi$ [3]. Thus $L = 1$. The strict decrease makes the first term the attained maximum, proving the sharp upper bound as well.

Follow-up tasks after the full solution

Task 2. Obtain explicit bounds for $q_k - 1$ with optimal leading constants and rigorous remainders uniform in k.

Task 3. Determine which exponents p make the ratios of consecutive zero spacings of $y'' + \kappa z^{-p} y = 0$ decrease, remain constant, or increase, including the relevant singular endpoint conditions.

Task 4. Classify the real orders ν for which the positive gaps $j_{\nu,k}^2 - j_{\nu,k-1}^2$ are log-concave, specifying separately any adjoined zero at $k = 0$.

Task 5. Use the integral balance (4) to test whether the analogous affine comparison closes the LGL interval-ratio conjecture in the same source, including the dependence on polynomial degree.

Verification and status

The accompanying verifier checks the differential equation, affine transformation, potential difference, Wronskian identity, and gap-ratio algebra symbolically. Supplementary zero calculations test the strict inequalities and the location of the potential crossing; they are not the proof of the infinite assertions.

MathDB was checked live before the attempt and again before registration: the record was open and its claim fields were empty. The original source, the author's 2018 thesis explicitly retaining the exact conjecture, both indexed citing works, and public claim catalogues were checked. The literature audit records the sources, dates, and access limits.

References

[1] K. Brix, C. Canuto, and W. Dahmen, *Nested dyadic grids associated with Legendre–Gauss–Lobatto grids*, Numerische Mathematik **131** (2015), 205–239, doi:10.1007/s00211-014-0691-4. Author preprint: *Legendre-Gauss-Lobatto grids and associated nested dyadic grids*, arXiv:1311.0028v1 (2013).

[2] K. Brix, *Robust Preconditioners for hp-Discontinuous Galerkin Discretizations for Elliptic Problems*, doctoral thesis, RWTH Aachen, 2018; repository publication 2019, doi:10.18154/RWTH-2019-10662. Conjecture 3.36, p. 38; its reference [34] is the final journal paper above. `https://publications.rwth-aachen.de/record/772363`.

[3] NIST Digital Library of Mathematical Functions, §10.21(vi), equation (10.21.19), `https://dlmf.nist.gov/10.21#vi`.